2-4-26

TUTORIAL - 11

Age	Income	credit Score	Actual class	Pred Prob	Pred output
25	40	650	1	0.90	1 (TP)
30	60	700	1	0.75	1 (TP)
45	35	600	0	0.60	1 (FP)
50	30	580	0	0.40	0 (TN)
28	55	720	1	0.30	0 (FN)

class 1 → Positive

class 0 → Negative

1) Threshold value = 0.5

Prob > 0.5 → 1

Prob < 0.5 → class 0

	Actual 1	Actual 0
Pred 1	TP 2	FP 1
Pred 0	FN 1	TN 1

$$\text{precision} = \frac{TP}{TP+FP} = \frac{2}{2+1} = \frac{2}{3} = 0.67$$

$$\text{Accuracy} = \frac{TP+TN}{N} = \frac{2+1}{5} = \frac{3}{5} = 0.6$$

$$\text{Recall (TPR)} = \frac{TP}{TP+FN} = \frac{2}{2+1} = \frac{2}{3} = 0.67$$

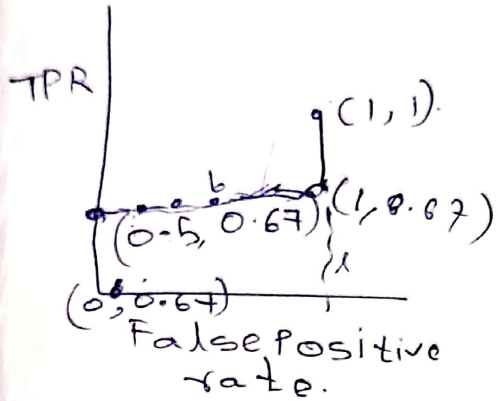
$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot 0.67 \cdot 0.67}{1.34}$$

$$= 0.67$$

Scanned with OKEN Scanner

False Positive Rate (FPR) $= \frac{FP}{FP+TN} = \frac{1}{1+1} = 0.5$

ROC curve



2) Threshold = 0.7

TP 2	FP 0
FN 1	TN 2

Prob > 0.7
prob < 0.7
TPR = 0.67
FPR = 0.0 (0, 0.67)

Threshold = 0.6 prob

TP 2	FP 1
FN 1	TN 1

TPR = 0.67, ~~0.5~~
FPR = 0.5
(0.67, 0.5)

Threshold = 0.5

TP 2	FP 1
FN 1	TN 1

threshold = 0.4

TP 2	FP 1
FN 1	TN 1

threshold = 0.3.

3	2
0	0

TPR = 1
FPR = 1
(1, 1)

Scanned with OKEN Scanner